T R U T H C H E C K

Fact-checking Health Claims on YouTube.

An LLM agent that distinguishes settled medical consensus from genuine scientific debate with citations.

Final presentation

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OVERVIEW

Why a fact-checker for health videos

THE PROBLEM

- Claims made by YouTube videos might be misleading
- Need to fact check to decide whether a video is legit or not

- Fact check should be done with authoritative sources like:

PubMed
peer-reviewed studies, via E-utilities API

Cochrane (through PubMed)
systematic reviews and meta-analyses

CDC / WHO / NIH
official guidance and consensus statements

APPROACH

An agent, not a classifier.

Given a YouTube URL:

- TruthCheck extracts every checkable health claim
- Retrieves real peer-reviewed evidence
- Classifies each claim against scientific consensus

THE DIFFERENTIATOR

Clear consensus gets a verdict.

Active scientific debate gets both sides.

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DATA COLLECTION

Six videos, one preprocessing finding – included in midterm

The corpus is clean and on-topic — the only real signal of trouble is caption punctuation on auto-generated transcripts.

6 videos

Huberman Lab + Nutrition Made Simple. ~72k words, 6 hours of long-form health content.

100%

FIT 1M CONTEXT WINDOW

Largest video: ~29k tokens. No chunking needed for Gemini 2.5 Pro.

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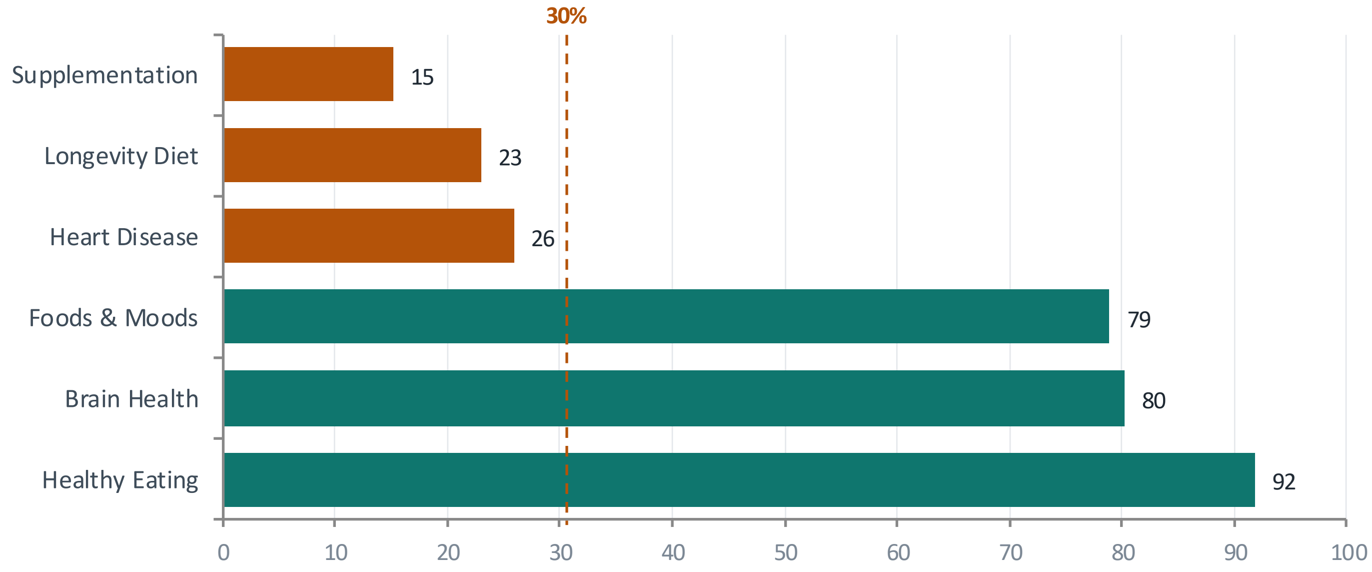
POORLY PUNCTUATED

Auto-generated captions drop to 15–26% punctuated segments — preprocessing step required.

PUNCTUATION RATE BY VIDEO

Manual captions vs. auto-generated



LLM-readable threshold

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MODEL & EVALUATION

A 5-stage Gemini pipeline, grounded in real papers

Each stage is a focused LLM call or a fan-out to PubMed via NCBI E-utilities.

The stages don't talk to each other directly; they hand off objects.

Pydantic schemas form the contract between stages

STEP 1

Transcript

youtube-transcript-api

→

STEP 2

Claim extraction

Gemini 2.5 Pro

→

STEP 3

Evidence retrieval

PubMed · Cochrane · WHO

→

STEP 4

Consensus eval

Gemini, grounded

→

STEP 5

Report

JSON + Markdown

LIVE TEST — "WHAT, WHY & HOW OF HEALTHY EATING" · 2 CLAIMS · 42 SECONDS

Claim	Verdict	Why this is the right call
"Nutrition is a leading factor in disease risk."	Unverifiable	Retrieved GBD papers describe methodology, not rankings. Model correctly declined to overclaim.
"Lifestyle fundamentals prevent many medical problems."	Unverifiable	Retrieved WHO/NIH papers tangentially related. Model grounded, didn't hallucinate consensus.

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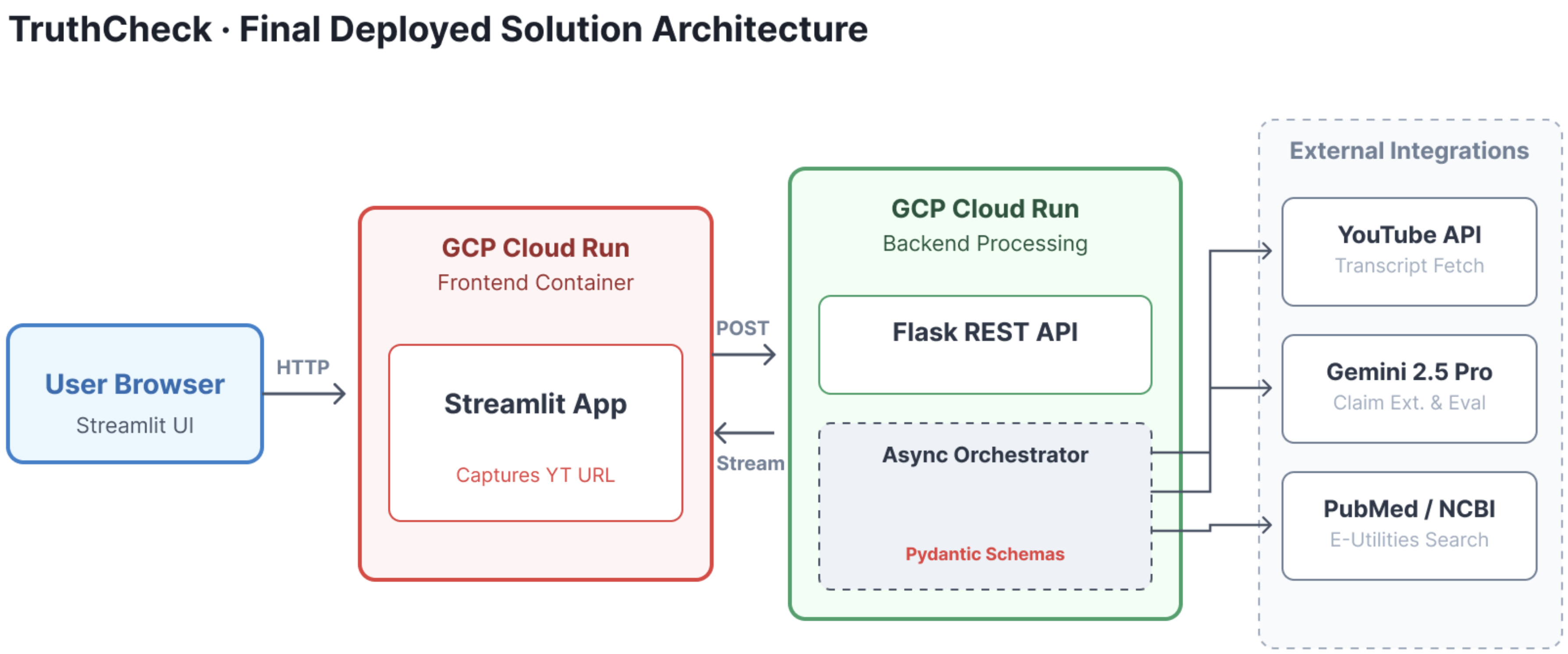
ARCHITECTURE

The deployed system, end to end



DEPLOYED ON Google Cloud Run · Docker images · scales to zero · no persistent database · GEMINI_API_KEY via env var

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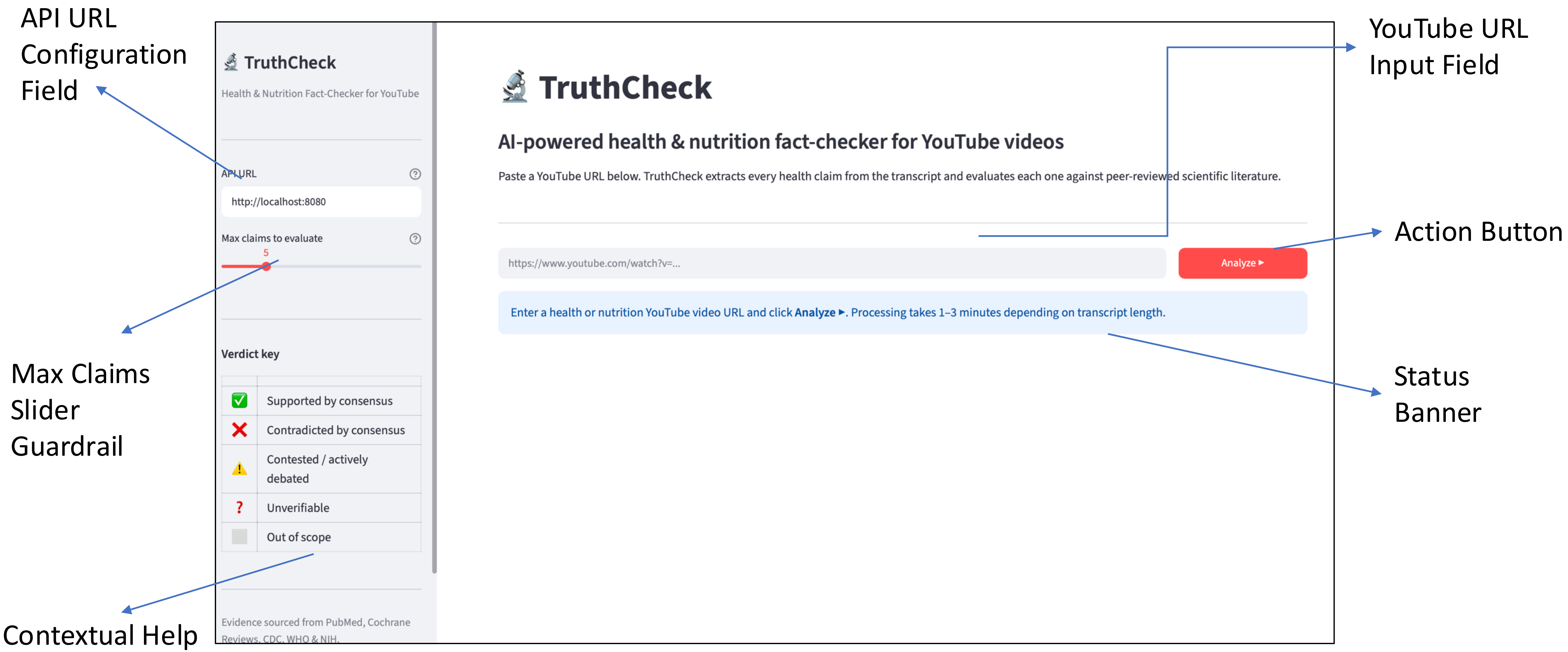
APPLICATION

What the user sees



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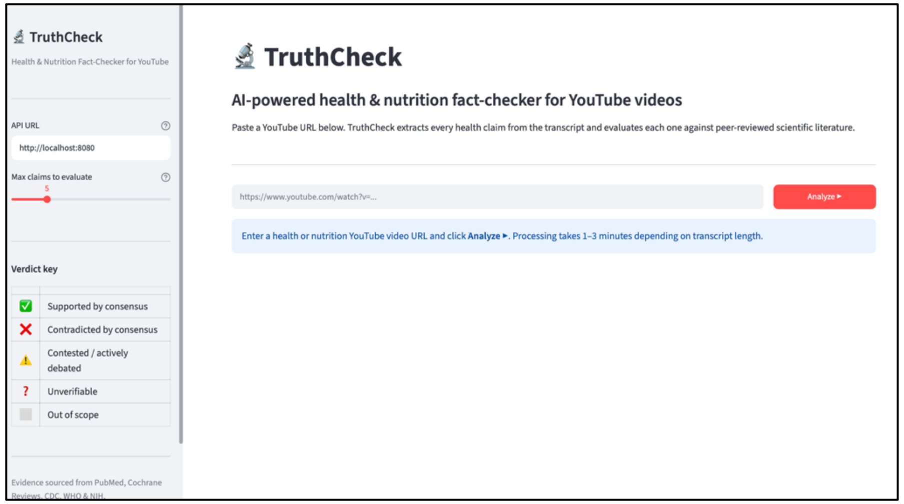
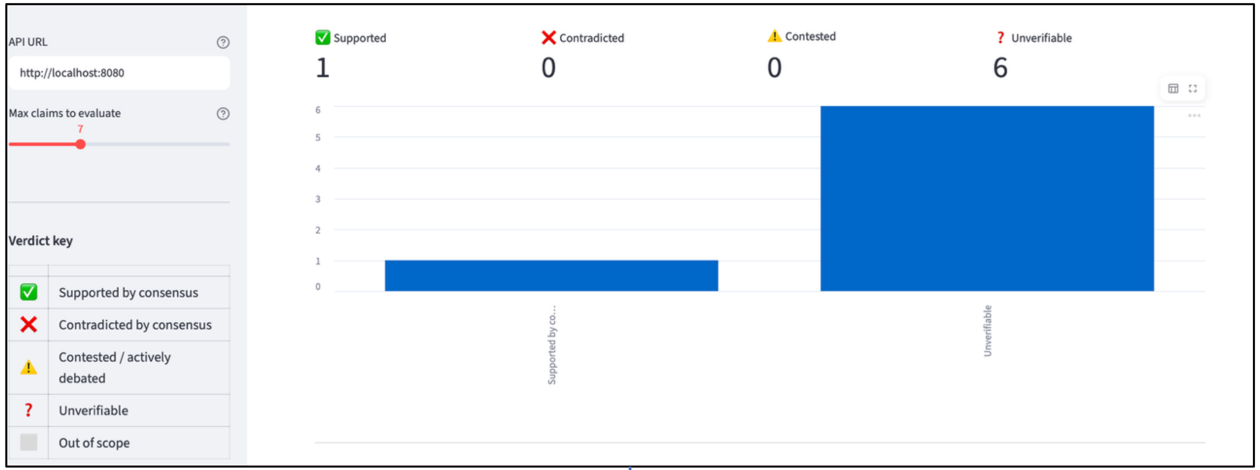
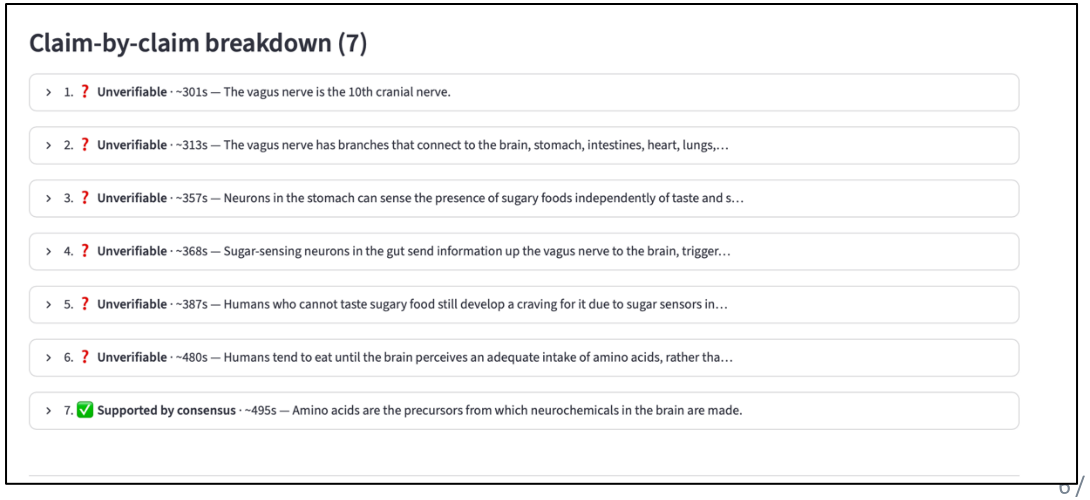
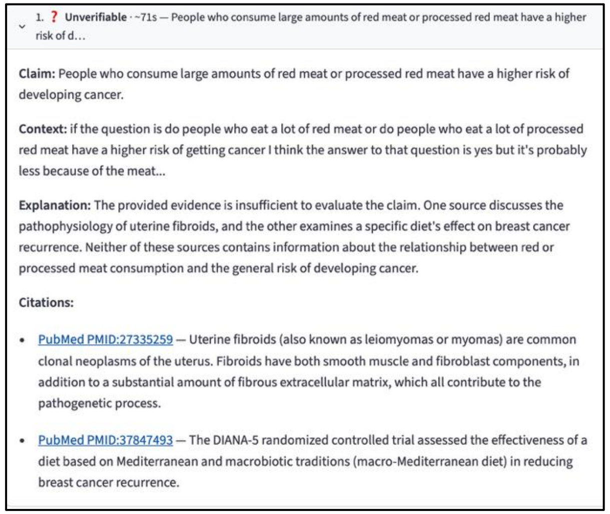
APPLICATION

Walk Through









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CHALLENGES & LESSONS

Three real problems I faced

Evaluation is a problem

PROBLEM

Evaluating multi-stage pipelines like TruthCheck is notoriously difficult because standard traditional software tests

FIX

Use LLM as a Judge or create a Golden Dataset

Scale Complexity of Long-Form Videos

PROBLEM

It generates hundreds of specific factual claims, which creates problems.

FIX

Max Claims Slider to cap total pipeline costs and making it easier to execute

Tangential evidence retrieval

PROBLEM

Verdicts skewed `unverifiable`.

FIX

Honest framing: the model correctly refuses to overclaim. Future work: query expansion + reranking.

THANK YOU!

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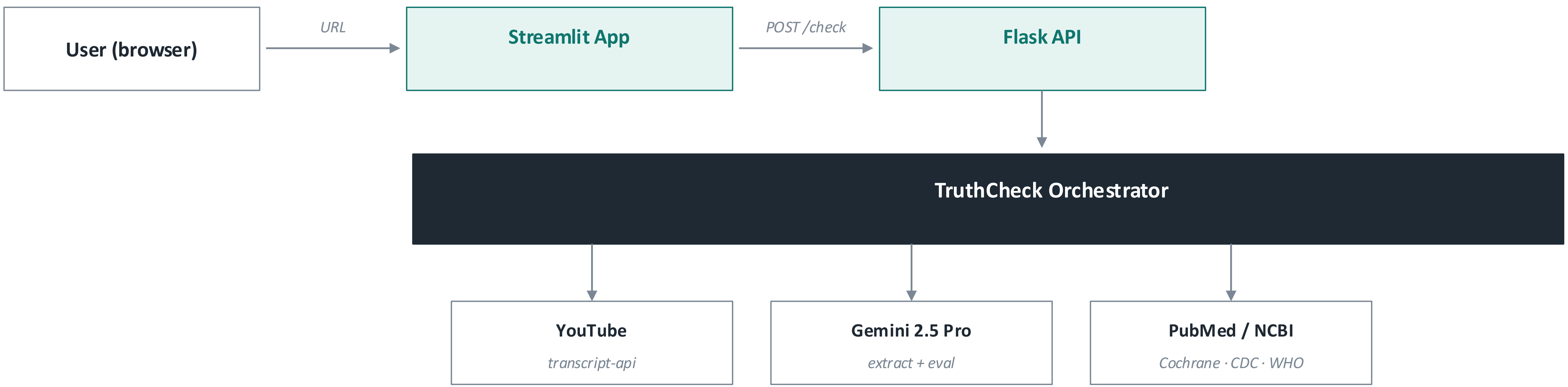
ARCHITECTURE

The deployed system, end to end

Two containerized services on Cloud Run: a Streamlit UI and a Flask REST API. Stateless, scales to zero, no database. The orchestrator calls Gemini twice (extract, evaluate) and PubMed once per claim.



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APPLICATION

What the user sees

USER FLOW

1 Paste YouTube URL

Plus a max-claims slider (1–20) to cap pipeline cost.

2 Watch progress live

Spinner with stage-by-stage messaging while Gemini runs.

3 Read the summary

4-metric row: supported · contradicted · contested · unverifiable.

4 Expand any claim

Timestamp, verdict, explanation, real citation URLs.

5 Download JSON / MD

Structured report for sharing or further analysis.